

GLOBAL HEALTH & LIFE EXPECTANCY ANALYSIS

An Interactive Power Bi Dashboard Using World Development Indicators

(2000–2024)

AnalystLab Africa Data Analytics Internship

Week 8 Capstone Project

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Dataset: World Development Indicators (WDI) World Bank

Analysis Tool

Microsoft Power BI

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1. Introduction

The World Development Indicators (WDI) dataset, published by the World Bank, is one of the world's most comprehensive sources of development statistics, covering over 200 countries and territories. The dataset contains indicators spanning health, education, economy, infrastructure, governance, technology, trade, poverty, and environmental sustainability.

This project focuses on analysing global health outcomes using selected health indicators from 2000 to 2024. The objective is to understand trends in life expectancy, maternal mortality, infant mortality, and healthcare expenditure while identifying relationships among these indicators. Microsoft Power BI was used to clean, transform, analyse, and visualise the data in the form of an interactive dashboard that supports informed decision-making.

The completed dashboard provides both an executive overview and a detailed analytical view, allowing users to explore health performance across countries through interactive filters, trend analysis, geographic visualisation, and comparative charts.

2. Objective

The primary objective of this project is to analyse global health indicators using the World Bank World Development Indicators dataset and develop an interactive Power BI dashboard that communicates meaningful insights.

Specific objectives include:

- Analyse global life expectancy trends between 2000 and 2024.
- Compare countries with the highest and lowest life expectancy.

- Examine maternal and infant mortality across countries.
- Investigate the relationship between healthcare expenditure and life expectancy.
- Develop interactive visualisations to support data-driven decision-making.
- Provide recommendations based on the findings.

3. Problem Statement

Health outcomes vary considerably across countries due to differences in healthcare investment, economic development, healthcare accessibility, and public health policies.

Despite improvements in global healthcare, many countries continue to experience low life expectancy alongside high maternal and infant mortality.

Understanding these disparities is important for governments, healthcare organisations, and development agencies seeking to improve health outcomes and allocate healthcare resources efficiently.

This project seeks to identify patterns, trends, and relationships among selected health indicators that can support evidence-based policy decisions.

4. Dataset Description

The dataset used for this project was obtained from the **World Bank World Development Indicators (WDI)** database.

Dataset characteristics include:

- Source: World Bank
- Time Period: 2000–2024
- Coverage: 264 countries and territories
- File Format: CSV

- Software Used: Microsoft Power BI

The analysis focused on four key indicators:

- Life expectancy at birth (years)
- Maternal mortality ratio (per 100,000 live births)
- Infant mortality rate (per 1,000 live births)
- Current health expenditure (% of GDP)

These indicators were selected because they collectively provide a broad measure of national health outcomes and healthcare investment.

5. Data Cleaning and Transformation

The raw dataset contained numerous development indicators covering multiple sectors. Since the project focused specifically on health outcomes, the data underwent several preprocessing steps in Power Query before analysis.

The following transformations were performed:

- Imported the World Development Indicators dataset into Power BI.
- Filtered the dataset to retain only four selected health indicators.
- Removed unnecessary columns not relevant to the analysis.
- Standardised column names for clarity and consistency.
- Converted columns to appropriate data types.
- Filtered and managed missing values where appropriate.
- Created a pivoted table to support multivariate scatter plot analysis.
- Created DAX measures for KPI calculations.
- Rounded KPI values to one decimal place for improved readability.

These steps ensured that the dataset was clean, consistent, and suitable for interactive reporting.

Analysis Methodology: The analysis followed the standard data analytics workflow:

Data Collection: The World Development Indicators dataset was downloaded from the World Bank.

Data Preparation: Power Query was used to clean and transform the dataset.

Data Modelling: Relationships were established within Power BI, and DAX measures were created to calculate key performance indicators.

Data Visualisation: Interactive dashboards were developed using:

- KPI Cards
- Line Charts
- Bar Charts
- Bubble Map
- Scatter Plot
- Slicers
- Text-based Insights
- Recommendations

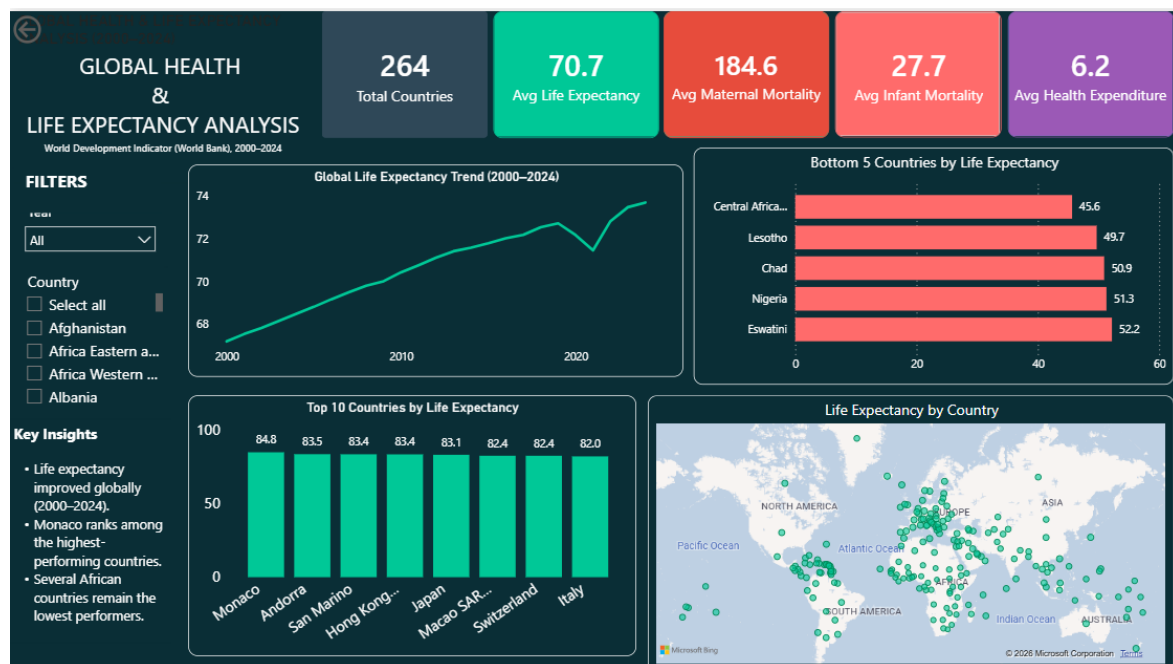
Interpretation: Patterns, trends, and relationships observed in the dashboard were interpreted to generate business insights and recommendations.

Dashboard Overview: The dashboard consists of **two interactive pages** designed to provide both a high-level executive summary and a detailed analytical perspective.

Page One – Executive Overview

The first dashboard presents an overview of global health performance through:

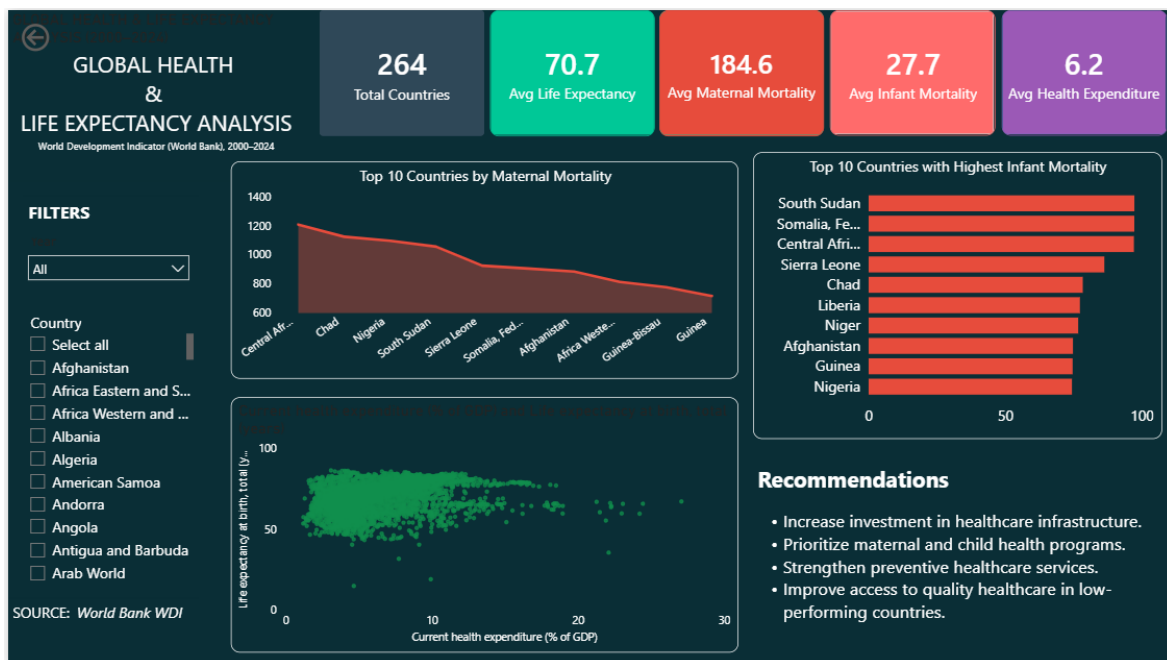
- Total Countries
- Average Life Expectancy
- Average Maternal Mortality
- Average Infant Mortality
- Average Health Expenditure
- Global Life Expectancy Trend
- Bottom Five Countries by Life Expectancy
- Top Ten Countries by Life Expectancy
- Global Life Expectancy Map
- Interactive Country and Year Filters
- Key Insights Panel



Page Two – Health Outcomes Analysis

The second dashboard focuses on relationships among health indicators through:

- Maternal Mortality Analysis
- Infant Mortality Rankings
- Scatter Plot showing the relationship between Health Expenditure and Life Expectancy
- Interactive Filters
- Recommendations Panel



Key Findings

The analysis produced several important findings.

- Global life expectancy increased steadily between 2000 and 2024, indicating overall improvements in healthcare and living conditions.
- Monaco, Andorra, Japan, Switzerland, and other developed nations recorded the highest life expectancy values.

- Several African countries recorded the lowest life expectancy, reflecting persistent healthcare challenges.
- Countries with high maternal mortality generally also experienced high infant mortality.
- A positive relationship exists between healthcare expenditure and life expectancy, although the strength of this relationship varies across countries.
- Geographic disparities remain evident, with developed economies generally outperforming developing nations on key health indicators.

Insights

The dashboard highlights significant inequalities in global health outcomes. Countries investing more in healthcare tend to achieve longer life expectancy and lower mortality rates, although healthcare spending alone does not fully explain health performance.

Factors such as governance, healthcare accessibility, education, and socioeconomic conditions also contribute to national health outcomes.

The analysis further demonstrates that maternal and infant mortality remain concentrated in lower-performing regions, suggesting the need for targeted interventions and sustained investment in maternal and child healthcare services.

Recommendations

Based on the findings, the following recommendations are proposed:

- Increase investment in healthcare infrastructure and public health systems.
- Strengthen maternal and child healthcare programmes in countries with high mortality rates.
- Improve access to quality healthcare services in underserved communities.

- Promote preventive healthcare and public health awareness campaigns.
- Encourage governments to use data-driven approaches for healthcare planning and policy formulation.
- Monitor health indicators continuously to evaluate progress toward sustainable development goals.

Conclusion

This project demonstrates how data analytics can support evidence-based decision-making in the healthcare sector. By analysing selected indicators from the World Bank World Development Indicators dataset, meaningful trends and relationships were identified across 264 countries between 2000 and 2024.

The interactive Power BI dashboard provides stakeholders with an accessible platform for exploring global health performance through KPIs, trend analysis, comparative rankings, geographic visualisation, and relationship analysis. The findings reinforce the importance of healthcare investment, particularly in improving maternal and child health outcomes and reducing disparities between countries.

Overall, this project illustrates the value of data analytics in transforming complex global datasets into actionable insights that can inform policy, planning, and strategic decision-making.

References

World Bank. (2025). *World Development Indicators*.

<https://datatopics.worldbank.org/world-development-indicators/>

Microsoft. (2025). *Microsoft Power BI Documentation*. <https://learn.microsoft.com/power-bi/>